

2. Show that the force on a charged particle with charge q and mass m undergoing uniform circular motion when traveling perpendicular to a magnetic field B is

$$\vec{F}_B = \frac{q^2 B^2 R}{m}$$

3. Derive the equation for the amount of charge Q going through a circular loop with radius R , experiencing a torque T from a perpendicular magnetic field B over a time t . What happens to the charge as time goes on assuming all other variables remain constant. Does your result make sense?